

UST-YUDOMA, Russian Federation

WMO: 310540

Lat: 59.183N

Lon: 135.150E

Elev: 697

StdP: 14.33

Time Zone: 9.00 (E09)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-56.8	-53.5	-59.5	0.2	-56.1	-56.5	0.2	-53.3	9.3	-22.0	7.5	-24.4	0.1	90	0.669

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	23.1	86.2	66.3	82.5	64.9	78.7	63.0	69.7	80.2	67.0	78.5	64.7	75.8	3.5	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
65.9	98.1	74.8	62.9	88.1	71.4	60.5	80.8	69.3	34.2	80.0	32.0	78.9	30.2	75.7	89.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 12.3	11.0	8.1	-59.9	91.9	3.5	3.0	-62.4	94.1	-64.5	95.9	-66.5	97.6	-69.1	99.8		
(5)			-60.0	75.3	3.5	5.5	-62.6	79.2	-64.7	82.4	-66.7	85.5	-69.2	89.5		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	18.1	-34.7	-24.3	0.1	26.3	46.4	60.0	65.1	59.1	43.6	21.0	-13.7	-34.3		
	DBStd	37.78	10.91	11.32	12.78	10.44	6.96	6.66	6.41	6.68	7.88	11.51	15.91	11.89		
	HDD50	12775	2624	2080	1548	710	156	7	0	7	221	899	1910	2613		
	HDD65	17247	3089	2500	2013	1160	578	176	80	207	642	1364	2360	3078		
	CDD50	1138	0	0	0	0	43	306	470	290	29	0	0	0		
	CDD65	133	0	0	0	0	0	25	84	24	0	0	0	0		
	CDH74	1955	0	0	0	0	22	516	1061	352	4	0	0	0		
	CDH80	578	0	0	0	0	2	128	354	94	0	0	0	0		
	WSAvg	1.9	0.7	1.0	2.1	2.7	3.3	2.6	2.2	2.2	2.4	2.2	1.2	0.6		
	PrecAvg	13.9	0.6	0.4	0.4	0.5	1.1	1.5	2.1	2.0	2.0	1.3	1.1	0.6		
(15) Precipitation	PrecMax	19.8	1.1	0.7	2.4	2.1	2.3	4.0	5.6	10.2	6.4	3.1	2.5	1.7		
	PrecMin	8.5	0.2	0.1	0.1	0.0	0.2	0.2	0.5	0.1	0.2	0.4	0.3	0.2		
	PrecStd	3.0	0.2	0.2	0.5	0.4	0.6	0.9	1.0	2.0	1.3	0.7	0.6	0.3		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	2.6	14.5	40.4	58.0	77.1	88.5	91.9	87.4	73.5	47.2	32.1	-1.8		
		MCWB	1.9	12.8	31.9	44.6	55.8	66.4	69.1	66.6	59.7	39.8	29.8	-2.3		
	2%	DB	-8.7	5.3	33.5	50.5	71.1	84.1	87.8	82.3	67.2	42.4	25.5	-6.5		
		MCWB	-9.1	4.4	27.5	39.5	53.1	63.7	67.7	66.6	55.5	38.4	24.2	-7.1		
	5%	DB	-14.5	-1.2	27.9	46.8	66.2	80.3	84.3	77.6	62.1	38.4	16.0	-11.8		
		MCWB	-14.8	-1.9	23.5	37.2	50.5	61.9	65.9	63.9	53.1	35.0	14.9	-12.2		
	10%	DB	-19.1	-7.2	22.6	43.0	62.1	76.5	80.3	73.4	57.8	35.2	8.7	-16.9		
		MCWB	-19.3	-7.7	19.3	35.0	48.2	59.8	65.0	61.9	50.7	32.7	7.8	-17.2		
	0.4%	WB	1.9	12.7	32.3	44.9	57.6	70.6	76.2	71.4	61.7	42.3	29.8	-2.1		
		MCDB	2.5	14.2	40.2	55.1	72.7	82.3	83.4	81.0	69.4	44.6	31.7	-1.8		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	-9.0	4.4	27.8	40.6	54.1	66.1	70.9	67.7	57.4	38.6	24.1	-6.9		
		MCDB	-8.6	5.2	33.2	49.2	67.8	79.9	81.7	79.0	64.8	42.2	25.6	-6.4		
	5%	WB	-14.8	-1.9	23.8	37.8	51.6	63.6	68.2	65.2	54.3	35.4	15.1	-12.1		
		MCDB	-14.4	-1.2	27.8	45.8	64.5	76.5	79.3	75.4	60.9	37.8	16.0	-11.8		
	10%	WB	-19.4	-7.7	19.2	35.4	48.9	61.4	66.1	62.7	51.5	32.9	7.9	-17.2		
		MCDB	-19.1	-7.2	22.6	42.5	60.3	73.2	77.7	71.4	56.5	35.0	8.7	-17.0		
(35) Mean Daily Temperature Range	5% DB	MDBR	13.7	22.3	30.8	25.5	22.6	24.9	23.1	22.3	19.0	13.9	15.4	11.4		
		MCDBR	16.8	25.4	32.8	27.0	32.0	32.8	31.3	30.4	28.7	15.0	16.8	13.6		
		MCWBR	16.5	24.6	26.8	17.6	17.1	16.2	14.9	15.7	17.8	10.9	15.8	13.0		
		MCDBR	16.8	25.3	31.0	25.2	28.8	29.0	25.6	27.3	24.8	13.9	16.9	12.9		
	5% WB	MCWBR	16.5	24.4	25.7	17.3	16.4	16.0	15.1	16.4	16.9	10.7	15.9	12.7		
(40) Clear-Sky Solar Irradiance	taub		0.233	0.266	0.313	0.397	0.448	0.431	0.497	0.434	0.348	0.307	0.244	0.209		
	taud		2.027	2.079	2.025	1.893	2.040	2.224	2.030	2.203	2.410	2.262	2.040	1.960		
	Ebn at Noon		179	235	255	249	246	255	232	239	248	218	168	142		
	Edh at Noon		16	26	37	51	48	41	48	38	26	22	15	11		
(44) All-Sky Solar Radiation	RadAvg		197	543	1094	1613	1652	1862	1702	1325	832	412	244	115		
	RadStd		10	20	37	79	147	143	117	122	59	54	22	5		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (3 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page